
BY 503: Cellular Fuel and Cellular Communication

Course Code	BY 503
Course Name	Cellular Fuel and Cellular Communication
L-T-P-C	3-0-0-3
Prerequisites	IC136 or faculty consent
Intended for	3rd and 4th UG, PG
Distribution	Elective
Approval	8th Senate (Oct 2015)

Course Contents

- **Module 1**

Cellular fate of nutrients metabolism: Glucose metabolism; Glucose transporters, Glycolysis, TCA cycle, glycogen synthesis, gluconeogenesis, and glycogenolysis. Metabolism of amino acids and proteins, Metabolism of lipids; oxidation of fatty acids, ketone bodies and ketosis, de novo synthesis of fatty acids, Metabolism of nucleic acids; Biosynthesis and breakdown of purine and pyrimidine nucleotides, Salvage pathways. [15 Lectures]

- **The cellular internet**

The essential elements of cellular transduction mechanisms that allow signaling from the cell surface to the nucleus; reception, transduction and response. Types of signals: Endocrine, Paracrine, Neural, and Juxtacrine. Receptors and receptor trafficking, Types of Cell surface receptors: G-protein coupled receptors, Receptor tyrosine kinase receptors, Cytokine receptors and Non-tyrosine kinase receptors, Integrin receptors, Toll-like receptors, Ligand gated ion-channels receptors, Receptors with other enzymatic activities. Second messengers; Type of secondary molecules; diacylglycerol, phosphatidylinositols, cAMP, cGMP, IP3, and Ca²⁺. [15 Lectures]

- **Hormone and Endocrine system**

Body's long distance regulator; Hormones, Local regulators, Neurotransmitters, Neurohormones, and Pheromones. Type of hormones, Major endocrine gland, and Hormone transport, Hormone receptors - cell surface and intracellular, Mechanisms of hormone action, Neuroendocrine interactions. [12 Lectures]

Text Books

1. Bruce Alberts, Alexander Johnson, Julian Lewis, Martin Raff, Keith Roberts, Peter Walter, **Molecular Biology of the Cell**, 5th edition.
2. David L. Nelson, Michael M. Cox, **Lehninger Principles of Biochemistry**, 6th edition.
3. Mac Hadley, Jon E. Levine, **Endocrinology** 6th Edition, Pearson Prentice Hall Publication (for the endocrinology portion).

References

1. Jane B. Reece, Lisa A. Urry, Michael L. Cain, Steven A. Wasserman, Peter V. Minorsky, Robert B. Jackson, **Campbell Biology**, 10th edition, Benjamin Cummings, 2013.